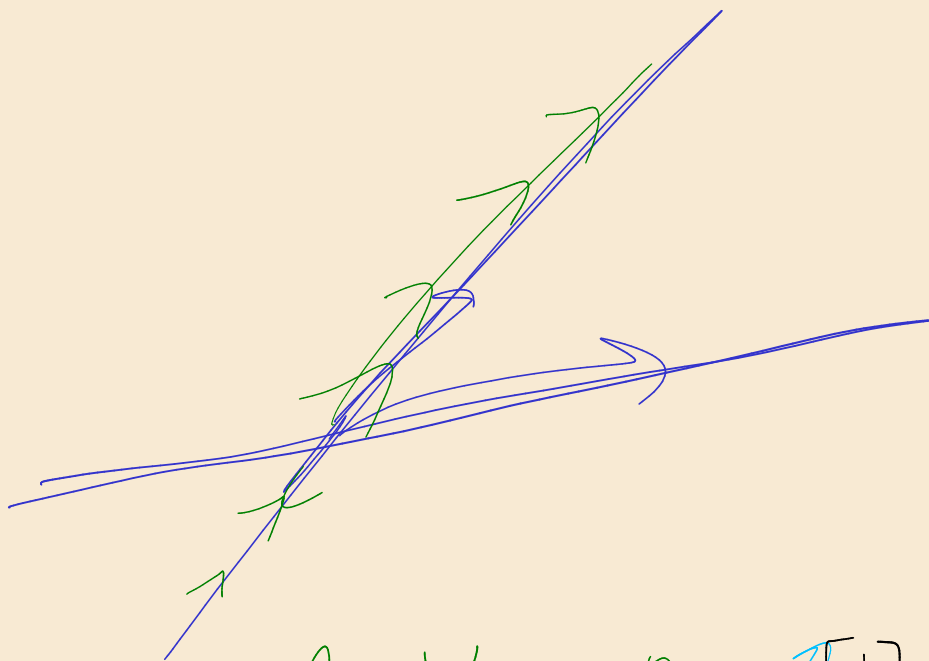
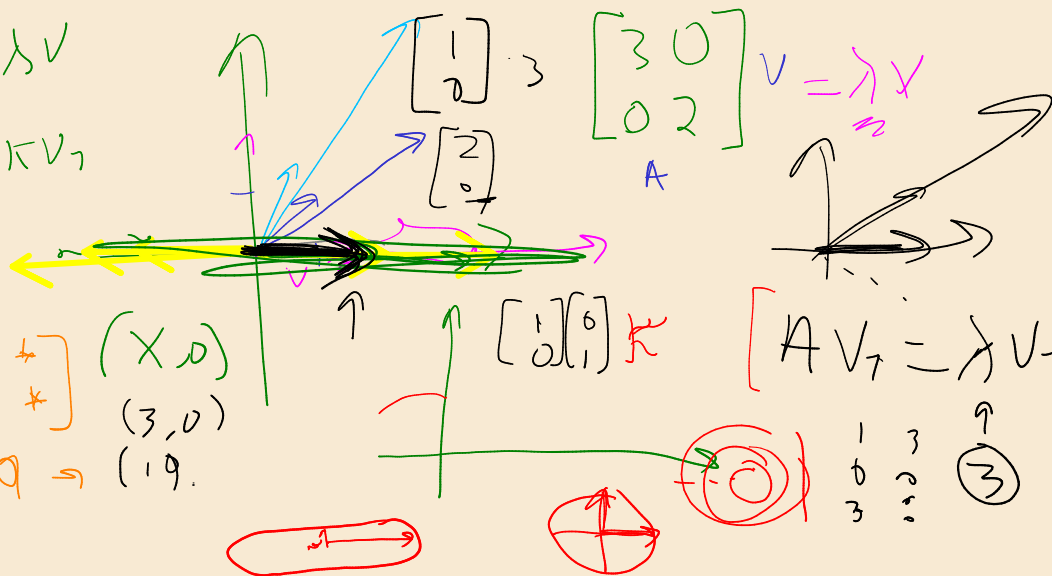




1. $U \subset V \subset S \subset W$
2. 6, 8, 9, 10, (11)
3. 2, 4, 6, 8, 10
4. $U \subset V \subset S \subset W$ (4, 3, 1)

$$AV_1 = \lambda V_1$$

$$V_2 = \kappa V_1$$



$$\begin{bmatrix} 2 & * \\ -1 & * \end{bmatrix} \begin{pmatrix} x \\ 0 \end{pmatrix}$$

$$(3, 0)$$

$$(1, 9)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \kappa$$

$$AV_1 = \lambda V_1$$

$$\begin{matrix} 1 & 3 \\ 0 & 3 \\ 3 & 3 \end{matrix}$$

🏠🏠🏠 06: Matrix subspaces and trace 🔗

Let $V = \{M \in M_{2 \times 2} : \text{tr}(M) = 0\}$ be the set of 2×2 matrices with zero trace.

1. Prove that V is a subspace of $M_{2 \times 2}$.
2. Find a basis and determine the dimension of V .

$\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid \text{tr}(M) = 0 \} \quad \forall c \in \mathbb{R}, M \in V$

$\forall v_1, v_2 \in V$
 $v_1 + v_2 \in V$

$$\begin{bmatrix} a_1 & b_1 \\ c_1 & -a_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 \\ c_2 & -a_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & -a_1 - a_2 \end{bmatrix}$$

$a_1 + a_1 = 0 \Rightarrow a_1 = -a_1$
 $a_1 + (-a_1) = 0$

E_1, E_2, E_3, E_4
 $E_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_4 = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$

$a(E_1 + E_4) \rightarrow a \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$c(a_1 + a_1) = c \cdot 0 = 0$

$V = \underline{c_1 a} + \underline{c_2 b}$

$c_1 \begin{bmatrix} x & y \\ 7 & x \end{bmatrix} + c_2 \dots$

$$a \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \end{bmatrix} \rightarrow$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\sqrt{11} = x_{11}$$

Unit Hypercube

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 3-0 \\ 0 & 2 \end{bmatrix} \quad \begin{pmatrix} 3-1 \\ 2-1 \end{pmatrix} = 0$$

$$\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \quad \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

08: Basis verification in different vector spaces

In each case, determine whether S is a basis for V :

a) $V = P_3, S = \{0, x, x^2, x^3\}$ $a+bx$

b) $V = M_{2 \times 2}, S = \left\{ I_2, \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \right\}$

c) $V = \mathbb{R}^3, S = \{e_1, e_2, e_1 + 3e_2\}$

d) $V = P_1, S = \{1-2x, 2-x\}$

$$a + bx + c(x^2 + 2x^3)$$

$$0 + bx + \dots$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\{0, 1, x, x^2, x^3\}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

$$M_{2 \times 2} \rightarrow$$

$$M_{2 \times 2} \sim \begin{bmatrix} a & b & c \\ a & b & c \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} 0 & 1 \\ 0 & 0 \end{matrix}$$

$$a(1-2x) + b(2-x) =$$

$$= a - 2ax + 2b - bx =$$

$$M_{2 \times 2} = \begin{matrix} a+2b \\ -x(2a+b) \end{matrix}$$

$$\boxed{503}$$

$$2a+b=0 \rightarrow b=-2a$$

$$a+2b$$

$$a-4a = \boxed{-3a} \quad a = -\frac{503}{3}$$

$$a(1-2x) + b(2-x) = 1 + 2x$$

$$\begin{matrix} a = \frac{4}{5} \\ b = \frac{1}{5} \end{matrix}$$

$$2, (v_1, v_2)$$

$$4 \quad (1, x, x^2, x^3)$$

$$2$$

$$1-2x \quad 2-x$$

$$\begin{aligned} -a(1-2x) &= (2-x) \\ (1-2x)a &= 2-x \end{aligned}$$

Suppose T is a linear transformation from $\mathbb{R}^2$ to P_2 such that:

$$T \begin{pmatrix} -1 \\ 1 \end{pmatrix} = x + 3, \quad T \begin{pmatrix} 2 \\ 3 \end{pmatrix} = 2x^2 - x$$

Find $T \begin{pmatrix} a \\ b \end{pmatrix}$ for arbitrary $a, b \in \mathbb{R}$.

$$T \begin{pmatrix} -1 \\ 1 \end{pmatrix} = x + 3$$

$$T \begin{pmatrix} a \\ b \end{pmatrix} =$$

$$\begin{aligned} T \begin{pmatrix} 2 \\ 3 \end{pmatrix} &= 2x^2 - x \\ c_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix}_{v_1} &+ c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}_{v_2} \end{aligned}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = c_1 v_1 + c_2 v_2$$

$$\underline{T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right)} = T(c_1 v_1 + c_2 v_2) =$$

$$= T(c_1 v_1) + T(c_2 v_2) =$$

$$= c_1 T(v_1) + c_2 T(v_2) \quad \begin{matrix} \nearrow \\ f(x) \end{matrix}$$

$$c_1 (x+3) + c_2 (2x^2-2)$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = c_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -c_1 + 2c_2 \\ c_1 + 3c_2 \end{bmatrix}$$

$$\underline{\underline{a+b = c_2}} \quad c_1, c_2$$

Without doing any calculations, determine the eigenvalues and eigenvectors for the following transformations by thinking about their geometric effects:

- a) **Shear matrix:** $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

- What vectors remain on the same line after this transformation?
- What vectors get stretched or shrunk (and by what factor)?

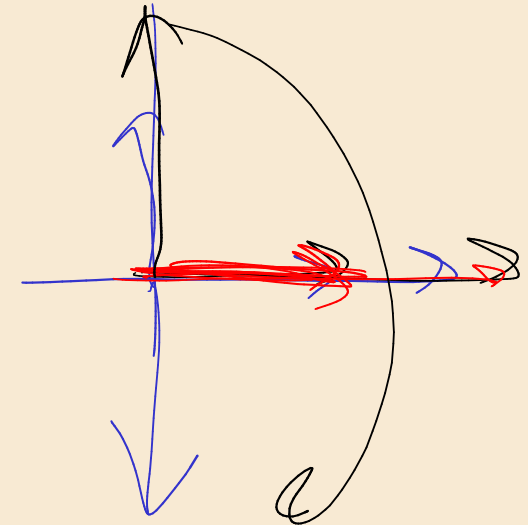
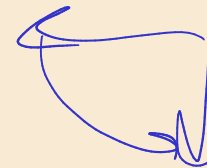
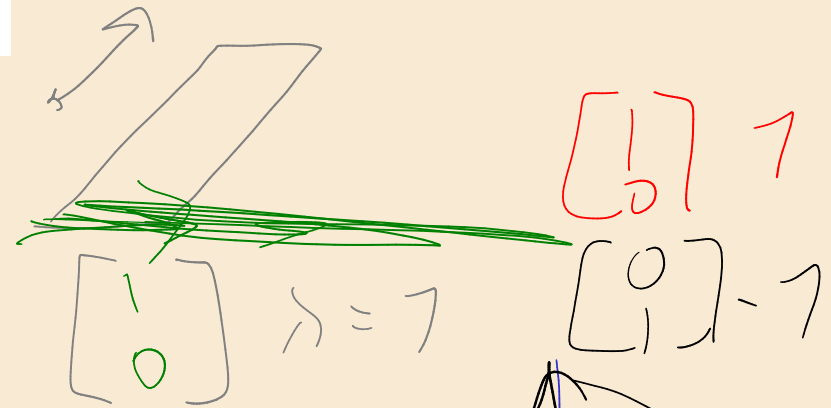
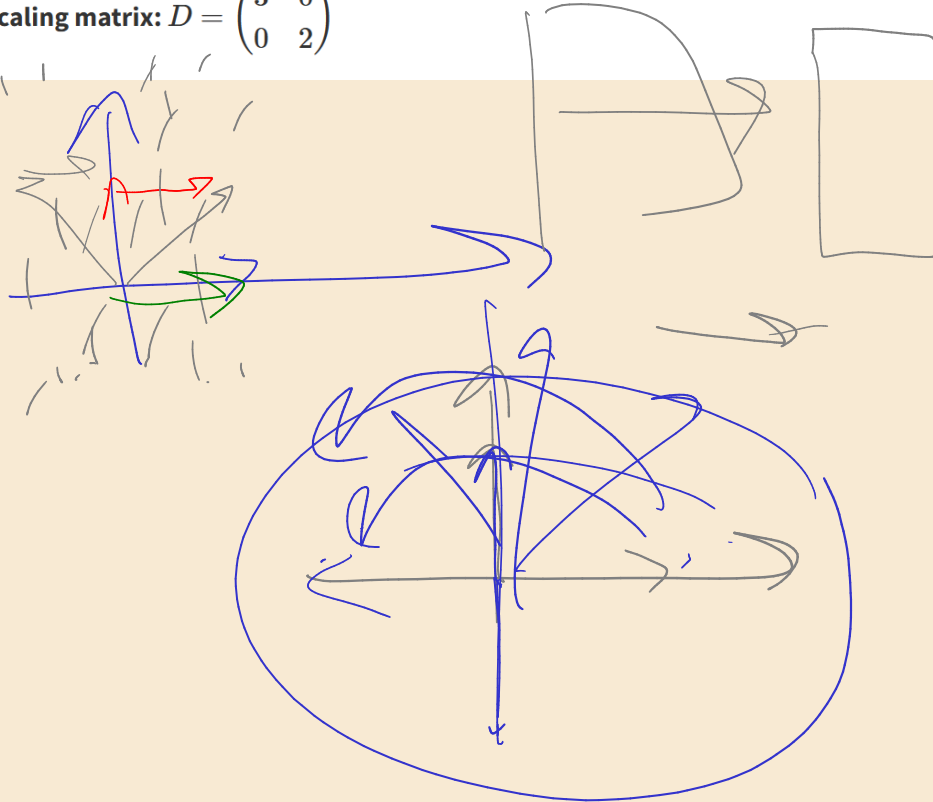
- b) **Rotation matrix:** $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (90° counterclockwise)

- c) **Reflection matrix:** $F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ (reflection across x-axis)

- d) **Scaling matrix:** $D = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$



Q,
J



$$\begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

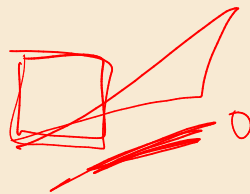
$$A v = \lambda v$$

$$A v - \lambda v = 0$$

$$v(A - \lambda I) = \underline{0}$$

$$|A - \lambda I| = 0$$

$$(A - \lambda I) \quad \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = 0$$



$$(2-\lambda)^2 - 1 = 0$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \vec{v}_1 = 3 \vec{v}_1$$

$$(2-\lambda-1)(2-\lambda+1) \\ (1-\lambda)(3-\lambda) = 0$$

$$\lambda_1 = 1 \\ \lambda_2 = 3$$

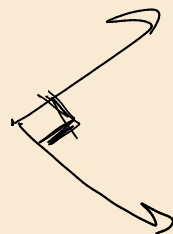
$$\begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix} \vec{v}_1 = 0$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \vec{v}_1 - 3 \vec{v}_1$$

$$\left[\begin{array}{cc|c} -1 & 1 & 0 \\ 1 & -1 & 0 \end{array} \right] \left[\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} - 3I \right] \vec{v}_1$$

$$-\cancel{x} + \cancel{x} = 0 \\ \cancel{x} - \cancel{x} = 0$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow 3$$



$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} x$$

$$x = y$$

